Practice Exam 2: Worked Solutions

Gym retention scenario · missed canceller = 4,000 · unnecessary offer = 500. Grade yourself only after a timed attempt.

Part A (12 p)

#	Answer	Concept
1	**C**	Leakage: the offer is sent only after the member has started to cancel. 0.74 → 0.97 is the signature.
2	**A**	Deployment fit: evidence must come from the population you will serve, including calibration.
3	**D**	Resampling inflates probabilities (ranking fine, size wrong). p* needs calibrated sizes.
4	**B**	Collinearity: separate coefficients unstable, prediction fine.
5	**C**	Confounding / sign flip (like students and balance).
6	**A**	Lower threshold → more flagged, recall up, precision down.
7	**D**	AUC does not choose the deployment; cost at the threshold, capacity and calibration do.
8	**B**	1 − 0.95^30 = 1 − 0.215 ≈ 0.79 (data snooping).

9 (2 p)

False. A small p-value shows an association in past data, not what happens if we change the input; members who visit more may simply be more motivated, so pushing visits up need not cut cancellations (only a randomized test shows that).

10 (2 p)

False. More rows reduce random sampling error but cannot bring back a group that was never collected; app members are a selection/representativeness gap that needs new data on them.

Part B (34 p)

B1 (3 p)

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z = −1.80 + 0.90·1 + (−0.30)·2 = −1.80 + 0.90 − 0.60 = −1.50
e^−1.50 = 0.223
p = 0.223 / 1.223 = 0.18
```

About **18%** chance of cancelling within 60 days.

B2 (5 p)

(a) e^−0.30 ≈ **0.74**: each extra visit per month multiplies the odds of cancelling by about 0.74 (about 26% lower odds), **holding contract type fixed**. Association, not cause.

(b) Different units: the contract term is a 0/1 switch, visits is per single visit. Three fewer visits per month (+0.90) has the same size as switching to a monthly contract, and across the observed range (0 to 20 visits) visits spans 6.0 on the log-odds scale. Compare realistic changes, not raw coefficients. Size is not cause either.

B3 (6 p)

(a) **Not supported.** The p-value shows that members who visit more cancel less, an association. Visits are likely a marker of motivation (confounding), and visits nudged by emails may not carry the same effect. The model predicts risk; it does not measure the effect of an email.

(b) Randomly assign at-risk members to **reminder emails vs no emails**. Compare the **60-day cancellation rate** (and visits) between groups. Random assignment makes the groups alike on motivation and everything else, so the difference is caused by the emails.

(c) Baseline balance; emails actually delivered and opened; cancellations recorded the same way for both groups; enough members; outcome fixed in advance.

B4 (4 p)

App members are **0% of the training data**: the model's scores for them are extrapolation, and they may behave differently (no branch tour, different motives). Calibration for them is unknown. **Limit:** do not use the model for app members yet; use a simple rule. **Learn:** score them in shadow mode, collect 60-day outcomes, then validate and recalibrate.

B5 (4 p)

(a) Among members with 0 to 20 visits per month, each additional visit per month is associated with **1.5 more units of monthly spend on extras** on average.

(b) Mean residuals are positive at both ends (+2.1, +2.6) and negative in the middle: a **U-shape**, so the relationship is curved and the straight line misfits (add a squared term or bands). The spread grows from 3.0 to 11.0, so predictions for heavy users are much less precise than the RSE of 8 suggests; only 50 members are in the top band.

(c) $12 + 1.5 \times 30 =$ **57**. Do not trust it: 30 visits is outside the data (0 to 20), so it is extrapolation, and the line is already off at the top of the range.

B6 (4 p)

(a) **Collinearity** ($r = 0.93$, $VIF \approx 1/(1 - 0.86) \approx 7.4$): both inputs carry the same information, so their separate effects cannot be told apart; SEs inflate (0.20 → 0.55, 0.30 → 0.80) while the fit barely changes (R^2 0.31).

(b) **No:** $0.75 \pm 2 \times 0.80 = -0.85$ to 2.35 includes 0, and "visits held fixed" barely occurs in the data.

(c) Prediction: keep the model with the best test performance; collinearity does little harm. Explanation: don't interpret the separate coefficients; use one input or an index, or collect data where they vary separately.

B7 (4 p)

(a) **Model 3:** training AUC 0.98 vs test 0.67, the largest gap and the worst test score (60 extra variables, high variance).

(b) **Model 2:** best test AUC (0.79), small gap (0.81 → 0.79), still few inputs to explain.

(c) **Who doesn't answer the survey.** 45% missing, and non-responders may be exactly the members about to cancel (missing not at random). Also check that the survey is available before the prediction date (no leakage) and that the gain lowers cost at the chosen threshold.

B8 (4 p)

(a) 0.90 ± 2 × 0.20 = **0.50 to 1.30**; odds ratio e^0.50 ≈ 1.65 to e^1.30 ≈ 3.67. Holding visits fixed, monthly-contract members plausibly have between about 1.6 and 3.7 times the odds of cancelling.

(b) The interval gives the **size and precision** (could be 1.6× or 3.7×); the p-value only says it is unlikely to be zero.

(c) **No.** With 2 million rows even tiny effects become "significant". e^−0.02 ≈ 0.98, only about 2% lower odds per visit: statistically significant but practically small. Judge importance by effect size and by whether it changes decisions or cost.

Part C (29 p)

120 cancelled, 880 did not.

Threshold	TP	FN	FP	TN	Recall	Precision	Specificity	Accuracy	Cost (4,000 / 500)
0.50	36	84	14	866	0.30	0.72	0.98	90.2%	343,000
0.20	78	42	102	778	0.65	0.43	0.88	85.6%	219,000
0.10	102	18	248	632	0.85	0.29	0.72	73.4%	196,000

C1 (5 p)

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Recall = 78/(78+42) = 78/120 = 0.65
Precision = 78/(78+102) = 78/180 = 0.43
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Recall 0.65: the offer reaches 65% of the members who would cancel; 42 of 120 are missed. Precision 0.43: about 4 in 10 offers go to members who really were about to cancel; the rest go to members who would have stayed.

C2 (7 p)

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(a) 4,000·18 + 500·248 = 72,000 + 124,000 = 196,000
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(b) **Model at 0.10**: lowest cost (196,000), 244,000 below offering everybody and 284,000 below offering nobody (if capacity allows; see C4).

(c) "Offer nobody" is 88% accurate while catching zero cancellers (88% of members stay anyway). Accuracy weighs a 4,000 miss like a 500 offer, so 90.2% hides that the 0.50 model still misses 84 of 120.

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(d) Saving vs best baseline (offer everybody 440,000) = 440,000 − 196,000 = 244,000
per 1,000
    Per year: 244,000 × 25 = 6,100,000
    Net: 6,100,000 − 2,000,000 = 4,100,000 per year (return ≈ 3×)
    Stop if running cost > 6,100,000 per year
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C3 (5 p)

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(a) p* = 500 / (500 + 4,000) = 0.111
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(b) Misses fall 84 → 18 (−66 × 4,000 = −264,000), unnecessary offers rise 14 → 248 (+234 × 500 = +117,000): net −147,000 (343,000 → 196,000). Total errors rise (98 → 266), so accuracy falls, but expensive errors are replaced by cheap ones. Accuracy counts, cost weighs.

C4 (7 p)

(a) **Feasible: 0.50 (50) and 0.20 (180).** 0.10 (350) is over capacity.

(b) **Choose 0.20.** It is feasible and cheaper than 0.50 at both ends of the range (135,000 vs 175,000; 387,000 vs 679,000), so it dominates. The mistake we can afford is an unnecessary offer (500); a missed canceller is the expensive one if its cost is near 8,000.

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(c) At 8,000: 387,000 − 268,000 = 119,000 per 1,000 members (170 extra contacts → ≈
700 per extra contact)
    At 2,000: 0.10 costs 160,000 > 135,000 → extra capacity would not help (0.20 is
already best)
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Worth asking for only if the true miss cost is near the high end; ask Finance to narrow the range first.

C5 (5 p)

Coordinates (FPR, TPR): 0.50 → (14/880 = 0.016, 0.30); 0.20 → (0.116, 0.65); 0.10 → (0.282, 0.85).

(a) **R = 0.50** (lowest, left), **P = 0.20** (middle), **Q = 0.10** (highest, right): a lower threshold moves the point up and right.

(b) The diagonal is a coin flip (TPR = FPR). AUC 0.81: in 81% of pairs, a randomly chosen canceller gets a higher score than a randomly chosen stayer.

(c) $p^* \approx 0.11$ → **Q (0.10)**, the cheapest point (196,000) if capacity allows; within 200 contacts, **P (0.20)**. ROC has no costs, no base rate, no capacity and no calibration, so it cannot pick the point alone.

Part D (25 p)

Item 1 (10 p)

Claim	Verdict	Reason
1 · R^2 0.58 → 58% of next year	**Not supported**	Training fit, optimistic; next year's members may differ. Needs R^2 on new data.
2 · PT session users cancelled less → free sessions cut cancellations	**Not supported**	Self-selection: members who book sessions are more committed. Randomize the free-session offer.
3 · 89% accurate → ready	**Not supported**	Offering nobody is 88% accurate. Need recall, cost vs baselines, calibration.
4 · At 0.20 flags 78 of 120	**Supported**	TP = 78 at 0.20; recall 0.65.
5 · AUC 0.81 → 81% of flagged cancel	**Not supported**	AUC is about ranking; the share of flagged who cancel is precision, 43% at 0.20.

Item 2 (7 p)

(a) **New branches:** actual 110,000 is about **2.4×** expected 45,000; existing branches are close (160,000 vs 150,000, about 7% off). The model under-predicts members it has barely seen.

(b) Calibration by score band for new-branch members on recent outcomes, plus how many of them were in the training data.

(c) If new-branch members cancel clearly more than predicted (e.g. scored 8%, actually 20%), **lower their threshold** (offer to more of them) or recalibrate their scores upward; within capacity, use a simple rule for them until recalibrated.

Item 3 (8 p)

JUDGMENT (35 words): The model ranks cancellation risk well (test AUC 0.81). Within the 200-contact limit, threshold 0.20 costs 219,000 per 1,000 members versus 480,000 doing nothing. It predicts risk; it does not prove the offer keeps members.

ACTION (31 words): From Monday, the retention-team manager contacts members scored ≥0.20 (about 180 per 1,000); a random 10% get no offer, to measure its effect. Monitor monthly: new-branch members' actual versus expected cancellations.